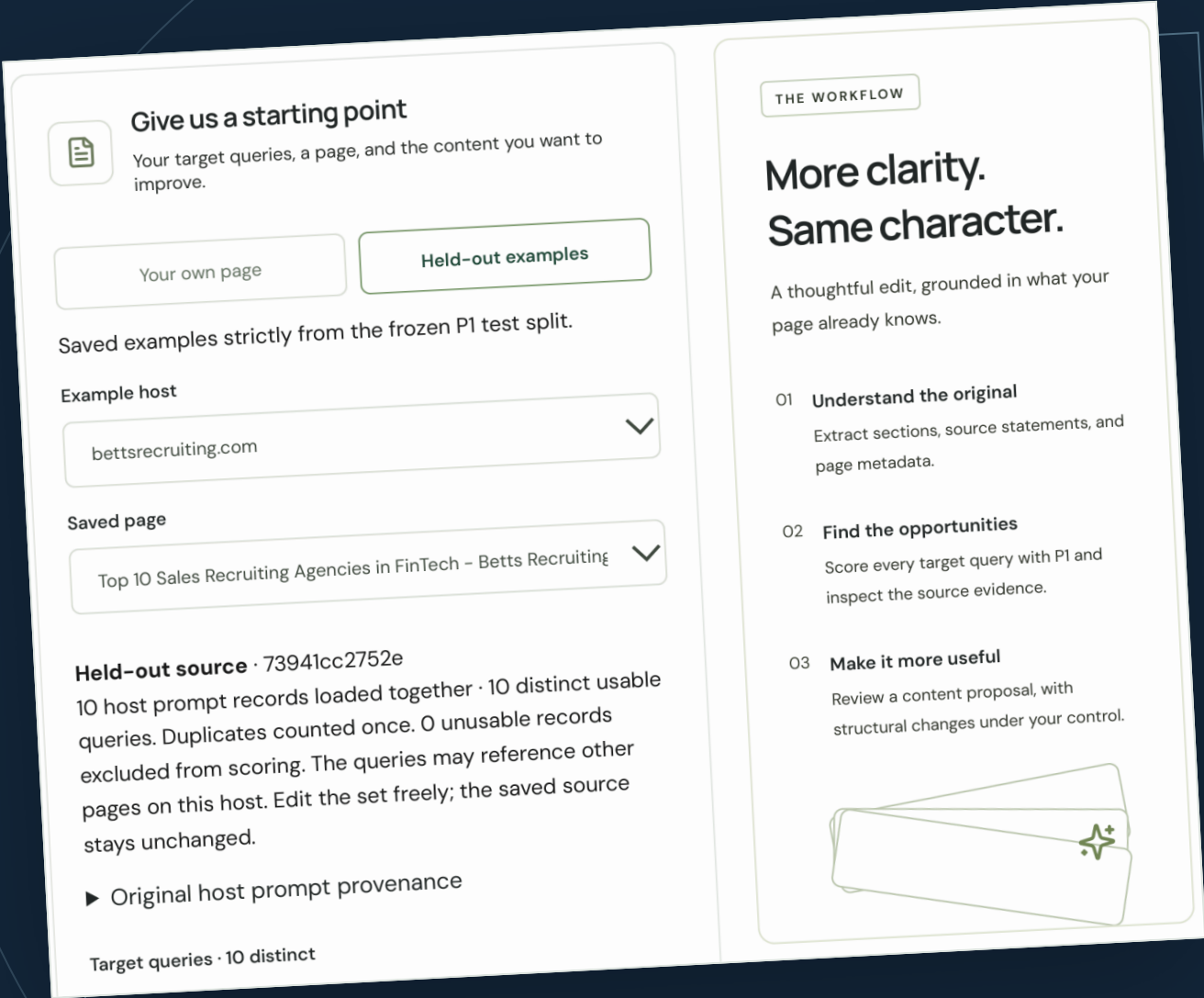


Multi-Query Content Optimizer

An explainable content engine that learns how to rewrite.

Analyze Explain Optimize

Shawn Lin / October 2, 2026



Content Studio · working prototype

Can a host's strongest pages teach us how to improve its weakest?

LEARN THE PATTERNS



Five strongest cited scenarios

Query alignment, document structure, and how the page answers the question.

APPLY THEM REPEATEDLY



Five weakest cited scenarios

A reusable optimization engine that improves supported content.

Commercial premise: lifting a customer's weakest pages creates value.

A strong, loosely held assumption to validate with publication outcomes and customers.

From a page and its queries to an explainable optimization loop.

SOURCE EXTRACTED

Top 10 Sales Recruiting Agencies in FinTech - Betts Recruiting

bettsrecruiting.com · Held-out example · 1674 words · 80 blocks

Structured HTML extraction

Parsed

Structured blocks and source references retained.

Edit source

Extraction checks

Consistency checks passed

78

blocks with text

0 list / quote containers have no own text

9

mappings need review

9 ambiguous · 0 unavailable

Inspect the source and score its query group.

OPENAI DRAFT · HUMAN REVIEW REQUIRED

A clearer starting point.

Held-out example · Review every change before it becomes your page.

Adjust direction

Export Markdown

Improved clarity and directness in describing assisted living with 24-hour staff and emergency support, options for caregiving without home placement in specific California locations, financial support and programs for dementia caregivers, payment methods for retirement communities, and home safety for parents with mild dementia in Calabasas and Berkeley. Information is aligned with the source and enhanced for easier user citation and use.

gpt-4.1-mini · gpt-4.1-mini--rewrite-page-v7 · 10681 input / 1208 output tokens · 19710 ms · Cost unavailable

Source-relative fidelity: rejected · LLM judgment; human review required.

START GEPA RUN

Run progress

Refresh runs

Saved runs

evidence-first-penalty-search-20261002-160122-ff558978 · running

▼

running · evidence-first-penalty-search-20261002-160122-ff558978 · working

Strategy contract: query-procedure-v1 · Edit boundary: body-content-v4 · Fidelity reviewer: gpt-5 (fidelity-slots-v4)

496/1000 dispatched rewrites · 0 reserved · 495 completed · 9 technical failures · 47 reflection calls

Usage totals reflect saved phases; in-flight provider usage appears when its response is saved.

Analyze

Source blocks + per-query P1

Supplied product captures. GEPA progress is an in-flight snapshot; completed-run results appear later.

Explain

Diffs + exact score effects

Optimize

Grounded rewrites + GEPA

03

Citation records: examples and data quality

9,700 records

970 hostnames

5 top + 5 bottom

Per host; both already cited

9,527 URLs

Distinct exact URL strings

9,011 prompts

Distinct nonblank strings

Two actual records • bryteflow.com

“Which is the best ETL tool to migrate data from SAP HANA to Redshift?”

LABEL	CITED PAGE	WORDS ¹	HEADINGS ¹
Top	Migrate SAP data to Redshift /sap-to-redshift	1,662	30
Bottom	SAP on AWS – The Fundamentals /sap-on-aws-fundamentals	3,921	9

A migration-specific page versus a broader fundamentals page—not a controlled before/after comparison.

¹ Exploratory text extraction. Source: part_001, rows 393 and 400.

Comparison limit: 920 / 970 hosts have no shared normalized nonblank query across labels.

Data-quality checks

91 empty prompts • 0.94%
70 top / 21 bottom; empty strings, not nulls.

28 excess exact duplicate rows
Separate from repeated URLs or prompts.

14 URLs have both labels
76 rows involved.

185 possible failed scrapes
Text-heuristic flags, not confirmed failures.

Same query, two substantive pages

TOPSample 01 · Matched examples

Which is the best ETL tool to migrate data from SAP HANA to Redshift?

Migrate SAP data to Redshift | BryteFlow

<https://bryteflow.com/sap-to-redshift>

Same query: migration-specific page versus broader fundamentals.

Extracted words	Headings
1,662	30
Title overlap	Payload chars
57%	134,111

► Read extracted text

► Inspect original row & raw payload

citations_part_001.parquet · row 393 · row numbering starts at 1

BOTTOMSample 02 · Matched examples

Which is the best ETL tool to migrate data from SAP HANA to Redshift?

SAP on AWS - The Fundamentals | BryteFlow

<https://bryteflow.com/sap-on-aws-fundamentals>

Same query: migration-specific page versus broader fundamentals.

Extracted words	Headings
3,921	9
Title overlap	Payload chars
14%	149,677

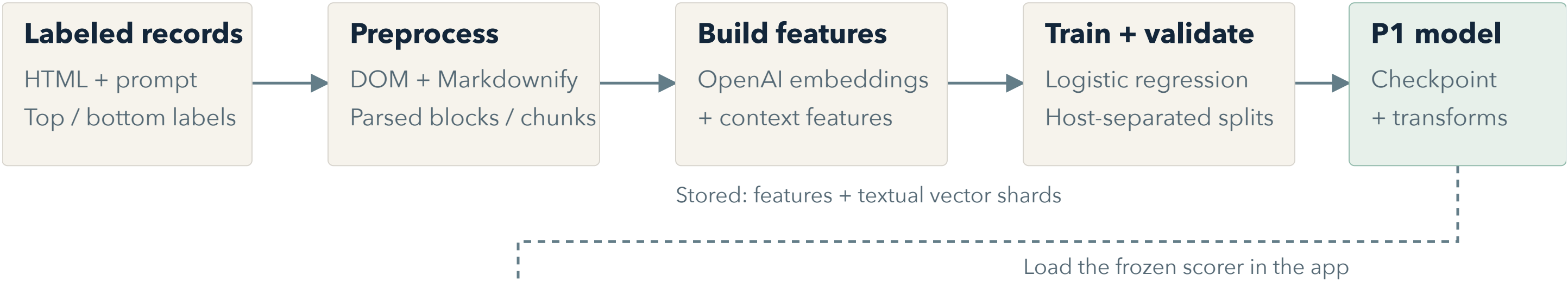
► Read extracted text

► Inspect original row & raw payload

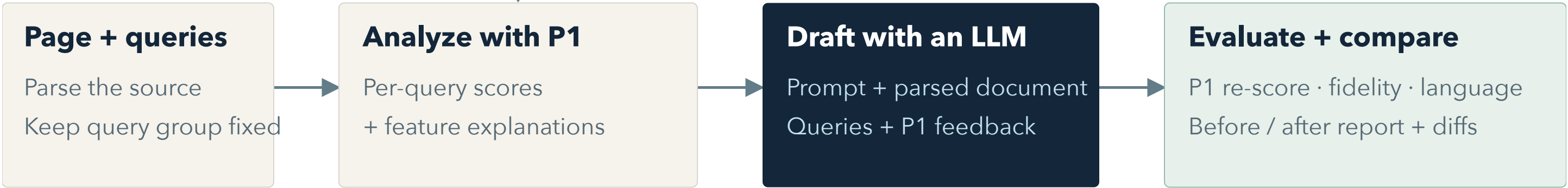
citations_part_001.parquet · row 400 · row numbering starts at 1

Train the scorer. Use it to guide a rewrite.

01 / OFFLINE SCORER TRAINING



02 / DEMO APP • REWRITE HARNESS



Shared preprocessing and feature contract for training, original-page scoring and draft re-scoring.

A source contract every stage can share.

Retain the evidence, preserve its structure, then score and rewrite the same representation.

REAL MARKDOWNIFY CORPUS RECORD • SELECTED FIELDS

```
block_id      "b000002"
type          "heading"
heading_level 1
text          "Top early-stage
               investors in the Nordics"
mapping_status "normalized_match"
source_locator DOM path to the H1
schema_version "dom-blocks-v3"
```

Blocks preserve identity and provenance. Chunks group evidence; derived embeddings represent the text.

Retention is a tradeoff

Earlier held-out HTML benchmark

METHOD	RETAINED	LEAKAGE
Baseline	121/129	30/50
Trafilatura	100/129	11/50
Readability	111/129	11/50
Conservative DOM	126/129	46/50

Selected anchors, AI-assisted references. This report predates the final Markdownify choice.

Final serving contract: Markdownify
3,072D textual embeddings in local vector shards.

P1 combines semantic alignment with page context

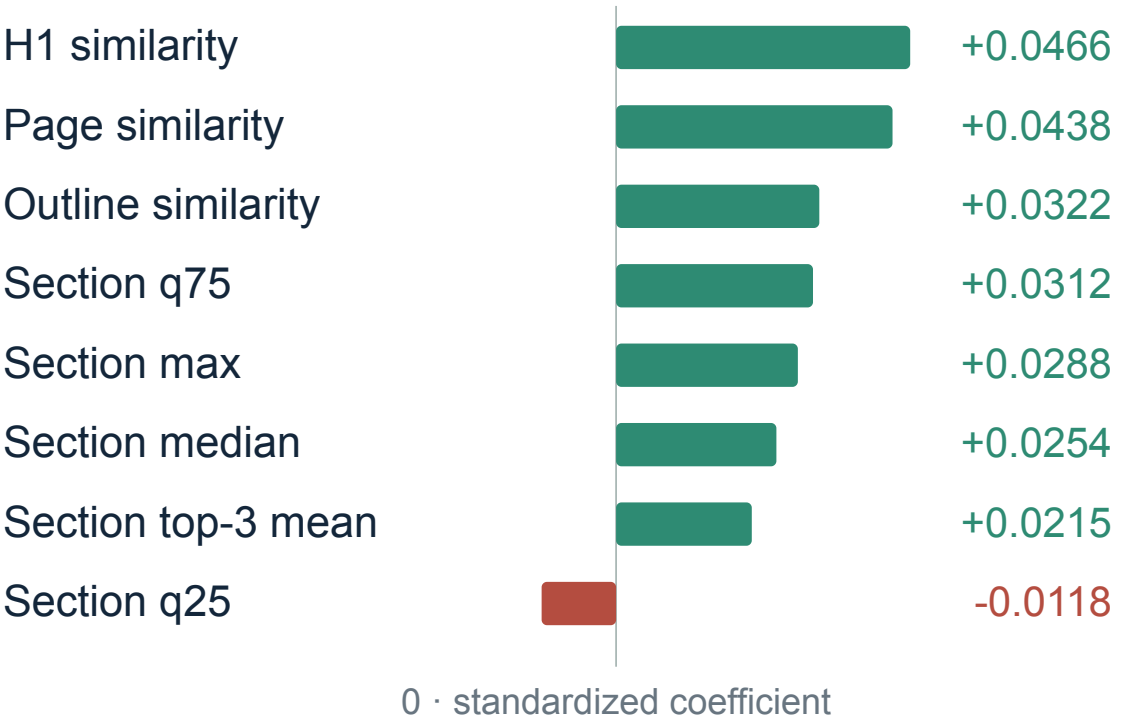
55 crafted inputs: 10 semantic similarities + 45 context features.

FEATURE FAMILY	COUNT	EXAMPLES
Semantic alignment	10	Query ↔ URL path / title / H1 / page; best-section and top-3 section similarity
Query characteristics	4	Prompt length; question / comparison / how-to flags
URL context	5	Path depth; homepage / editorial / commerce flags
Size & structure	13	Log word / heading counts; headings per 1,000 words; has table
Text composition	14	Unique-word / numeric-word fractions; paragraph length; links per 1,000 words
Source metadata	4	JSON-LD / article schema; source script count; empty title
Extraction quality	5	Retained-text fraction; sparse body; possible error response

v7.1 report ↗ · No hostname identity input. Counts describe raw features, before missingness indicators and scaling.

Semantic-similarity weights

From your saved weight report · page 3



Fitted weights, not editing importance.
Editing effect = weight × feature change.

Full weight report (pages 3-6) ↗ · All coefficients ↗

The score moves. The interface shows why.

+0.13

displayed P1 points
for this query

+0.0051

sum of log-odds effects

Page similarity helped.
Unique-word fraction offset some gain.
Every changed feature contributes.

The whole query group still regressed:
37.99 → 37.91

Target query	Before	After	Change
1. Where can I find assisted living with 24-hour staff and emergency support?	46.74	46.86	+0.13
Feature effects add to +0.0051 log odds. Probability changes are shown only for the complete model output.			
Page similarity			+0.0042
Unique word fraction			-0.0031
Section max			+0.0028
Retained text fraction			-0.0018
Section q75			+0.0016
Section top3 mean			+0.0010
Numeric word fraction			+0.0010
Paragraph log p90 words			-0.0007
Headings per 1000 words			-0.0003
Section median			+0.0003
Log word count			+0.0001
Paragraph word fraction			-0.0001
Other changed features			+0.0000

Real assisted-living example. The displayed +0.13 rounds the underlying probability difference.

We started with what P1 taught us.

The baseline directs supported improvements across the query group. Feature analysis shapes the search guidance.

BASELINE REWRITER · VERBATIM

“Use the whole-document P1 feedback to aim for a higher equal-weight mean while avoiding individual regressions.”

“Make substantive body improvements where supported, not merely title changes.”

rewrite-page-v7.txt · fixed source and output contract

P1 OBSERVATION → PERMITTED ACTION

H1 and outline alignment

Clarify the supported subject of existing editable headings.

Page and section alignment

Make existing answers and relationships more direct in body paragraphs.

Strong signals can be fixed

Title and URL explain scores but stay outside the edit boundary.

Promoted prompt structure

SYSTEM	Role + goal	Evidence-bound rewriting for a group of queries
	Promoted procedure	Map chunks → classify support → edit → check → report
	Fixed constraints	Same-chunk evidence · edit boundaries · language · security
USER JSON	Task + queries	Whole-page task · target queries · objective · tone
	Scoring context	Per-query P1 · procedure rationale · reflection provenance
	Source context	Metadata + chunks · editable blocks · read-only evidence

Required response: one proposal → each editable block is an edit or null

One page has to serve more than one question.

P1 · all target queries	lr-semantic-v7.1 classifier
10 distinct usable target queries. Each query has equal weight.	
Mean score: 45.04 / 100 · Lowest: 20.32	
Target query	P1 score / 100
1. Top recruiting firms for sales roles in the Fintech industry	63.05
2. leading Miami tech sales recruitment services	45.67
3. NY top recruitment agencies	42.45
4. top-rated recruiting agency SF	47.30
5. best tech sales recruiters Denver	52.29
6. How do I compare two offers when one includes equity and one does not	20.32
7. Recommended sales recruiters for infrastructure and data center startups	45.63
8. best agency for AE job placements	40.26
9. best tech sales staffing agencies in San Francisco	54.60
10. Recruiters that specialize in hiring strategic business development leaders	38.89
P1 v7.1 experimental classifier: sampled within-host top class among already-cited pages. Not citation probability, factual verification or causal uplift. Semantic and context inputs use the same Markdownify feature contract.	
text-embedding-3-large · 3072 dimensions · 32 cached inputs · 0 embedding calls · 0 reported tokens	

Betts Recruiting · ten distinct queries from the held-out example.

10 queries

One coordinated proposal.
Equal weight as the baseline.

The mean is not the whole result

The review interface retains every regression.

2. How can I get help without sending my parent to a home?. I'm based in Torrance.	36.78	36.75	-0.03 · regression
3. How can I get help without sending my parent to a home?. I'm based in Tustin.	37.02	36.92	-0.10 · regression
4. Is there financial support for dementia caregivers?. I'm based in Thousand Oaks.	35.31	35.26	-0.05 · regression
5. Are there programs that help cover dementia care costs?. I'm based in Costa Mesa.	38.91	38.73	-0.18 · regression
6. How do I pay for a retirement community?	36.80	36.60	-0.20 · regression
7. How do I keep my parent safe at home with mild dementia?. I'm based in Calabasas.	37.04	36.91	-0.13 · regression
8. How do I keep my parent safe at home with mild dementia?. I'm based in Berkely.	35.35	35.22	-0.14 · regression

The proxy rewarded a rewrite that changed the page’s language.

h2

b000079 · Original source mapping: exact

Industry-Specific Benefits: Tailored CDN Solutions for Media, Software, Gaming, and SaaS

► Original source location

paragraph

b000080 · Original source mapping: exact

不同的行业对CDN的需求各异。媒体企业需要无缓冲的高清直播；软件企业关注快速的更新分发；游戏行业则强调极低延迟以保障公平竞技；SaaS企业依赖稳定、高可用的API调用。优秀的CDN供应商（例如BlazingCDN）针对这些需求提供定制方案，确保卓越的性能和可靠性。

► Original source location

English page context; rewritten paragraph switches to Chinese.

HISTORICAL FAILURE · P1 v2

Target query	Before	After	Change
5. Best scalable content delivery network for large audiences	62.21	65.58	+3.37
6. How do users compare latency and performance across Azure, Cloudflare, and IBM solutions?	52.02	54.20	+2.18
7. Best content delivery network for product leadership	63.48	66.55	+3.06
8. top software dev tools	45.88	49.22	+3.34
9. CDNを活用してページ遅延を減らす方法	44.44	48.05	+3.61
Frozen P1 v2: sampled within-host top class among already-cited pages. Not citation probability, factual verification or causal uplift. Extraction-held-out does not imply P1-test-held-out.			

+3.61

P1 points on the Japanese query

A better score can still be an unacceptable edit.

Source-relative checks catch changes a score cannot judge.

Fidelity concerns — review this draft

The draft is retained for inspection and export. This automated assessment is advisory; review the source before using the changes.

b000043 · unsupported · Adds unsupported benefit/claim

The rewrite introduces “guidance and resources available in your area,” which is not stated in the source text; while the phone number and Eldercare Locator appear in a neighboring block, this specific benefit language is not supported by the original sentence.

Source references: None identified by the reviewer

b000054 · unsupported · Changed quantitative detail

The rewrite changes “often six bed facilities” to “typically with up to six beds,” introducing an upper limit not present in the source.

Source references: b000054

Actual review: an added local benefit and a changed facility-size claim.

“Often six bed facilities” became “typically with up to six beds.”

The judge flags the invented upper limit. The finding stays inspectable.

Mechanical contract

Valid IDs, allowed edit targets, same-chunk evidence and complete output.

Language preservation

Follow each original block’s language, even when the query uses another language.

Factual review

GPT-5 findings inform human review and GEPA’s penalties. The judge can be wrong.

Prompts are the offspring. Pages are the fitness tests.

— — — — —

PARENT PROCEDURE

Put supported answers first.



— — — — —

REFLECTED MUTATION

Preserve the qualifications
the parent omitted.

01 / PARENT SELECTION

Choose a potential ancestor

Sample a parent from the selection-page frontier, after pruning redundant tied coverage.



02 / MUTATION FEEDBACK

Learn where fitness breaks down

Rewrite sampled reflection pages. Feed source, queries, edits, raw P1 and factual findings to the reflection model.



03 / OFFSPRING & FITNESS

Create and screen an offspring

Validate the proposed prompt; allow one logged repair. Compare child and parent on the same sampled pages.



04 / SURVIVAL TEST

Test fitness across environments

Only a strict reflection-reward improvement advances to all 30 selection pages. Update scores and frontier.

The mutable genome is the editorial procedure. Security, source boundaries and output contracts stay fixed.

Fidelity findings become per-block penalties

Reward = mean P1 – penalties for edited blocks with fidelity concerns

Unsupported → −0.05

Unsupported claims, changed facts, lost material qualifiers or important omissions.

Uncertain → −0.02

The checker cannot establish sufficient support.

Supported → no fidelity penalty

Count affected blocks—not factual issues.

Three unsupported claims in one block: −0.05.

Unsupported + uncertain in one block: −0.05.

The penalty does not distinguish issue count or severity within a block.

Example: two flagged blocks → 0.05 + 0.02

P1 0.41 − 0.07 = **reward 0.34**

Run budget ↗ · 60 reflection / 30 selection pages

50 proposals · 1,000 attempts · test excluded

Completed export: evidence-first-penalty-search-20261002-160122-ff558978. Penalty weights are exploratory.

ILLUSTRATIVE GPT-5 RESPONSE · NOT A RUN LOG

```
{
  "edits": {
    "b000041": {
      "verdict": "unsupported",
      "category": "changed_quantitative_scope",
      "reason": "The source gives a typical size, not a maximum capacity.",
      "source_ids": ["b000041"]
    },
    "b000087": {
      "verdict": "uncertain",
      "category": "ambiguous_service_hours",
      "reason": "Extended hours does not establish evening coverage.",
      "source_ids": ["b000087"]
    }
  }
}
```

Schema-shaped excerpt; reasons wrapped for display. Each edit is judged only against its original chunk. Source IDs identify reviewed evidence, not proof of support.

The best raw score did not win.

The selected procedure balances relevance, factual deductions and execution reliability.

Prompt	Status	Raw mean P1 /100	Mean penalty /100	Mean reward /100	Pages with factual concerns	Execution failures
gpt-5-mini-conservative-chunk-aware-rewriting-map-queries-5279ea0e	evaluated · frontier	40.89	3.73	37.16	60.00%	0.00%
gpt-5-mini-new-procedure-enforces-strict-same-chunk-57c48649	evaluated · frontier	40.95	9.33	31.62	66.67%	3.33%

Supplied GEPA table · candidate 5279ea0e is the recommendation in the completed export.

FINAL SELECTION RESULTS · /100	RUN BASELINE	RECOMMENDED	CHANGE
Raw mean P1	40.8184	40.8921	+0.0737
Mean deduction	5.9333	3.7333	-2.2000
Mean reward	34.8851	37.1588	+2.2737

Both baseline and recommendation had 0% execution failures. Run baseline already contains a previously promoted procedure.

Evidence first became the procedure.

RECOMMENDED PROCEDURE · EXCERPT

“Conservative default: minimal single-paragraph edits, preserve original language/voice/qualifiers, and use harness fidelity feedback to remove unsupported changes.”

Saved editorial strategy · candidate 7dfc4cc5...

Map support before writing

Classify query support as full, partial or none against same-chunk evidence.

Keep the edit local

Preserve qualifiers and reuse supported wording. Report uncovered queries.

Promote with traceability

Versioned, model-specific procedure; saved evidence; explicit selection.

Rewriter model

gpt-5-mini

▼

Rewrite prompt

gpt-5-mini-conservative-evidence-first-rewriting-procedure-maps-59288bcd · selected default

▼

Prompts are specific to gpt-5-mini. Experimental prompts require review.

Reflection model

gpt-5-mini

▼

Product integration: a selected procedure becomes the seed for a later run. Capture shows the earlier promoted seed.

Test what actually improves the page

About 97% of the reward gain came from fewer fidelity penalties, not higher P1.
Compare four versions of the same pages to understand why.

1. Leave the page unchanged

- Keep the original words and layout.
- Use its score as the starting point.

3. Change only the structure

- Keep the words; change permitted formatting.
- Check whether P1 rises without a better answer.

2. Change only the wording

- Make existing answers clearer; add no new facts.
- Keep the headings, lists and layout the same.

4. Change both

- Combine clearer wording with better structure.
- Check whether the combination helps more.

For each version: compare P1 and factual errors. Ask reviewers which answer is more useful, without telling them how it was made.

Improve the value model. Then broaden the learning loop.

NEXT EXPERIMENT

Reward useful edits

Calibrate factual penalties and judge behavior.

Test a stronger P1 and expose actionable feature values.

Integrate the prototyped 0-3 usefulness judgments.

PRODUCT EVIDENCE

Learn from outcomes

Screen irrelevant query intents.

Add brand knowledge and competitor context.

Capture citation positions, co-occurrence and publication outcomes.

LONGER-TERM ROI

Make it cheaper to run

Distill strong-model traces into a smaller model.

Explore reinforcement fine-tuning, such as GRPO, after reward validation.

Keep promotion, monitoring and rollback explicit.

The next question: can supported, useful changes consistently improve this scorer?

Q&A